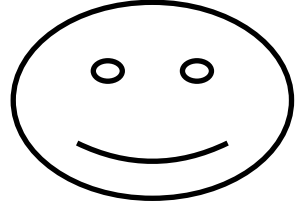


Networks

الشبكات



- what is networks ??

Is a group of connected computer and devices

اي (computer) او (devices) مشبوك مع بعض ضمن مجموعة واحدة

- for example :-

اجهزة المنزل كلها (home computers and devices)

. لمة بدي اضيف جهاز جديد فقط بعمل (join) للجهاز على (wi-fi) و هيك بكون عملت مشاركة للجهاز مع الشبكة ف صار ضمن مجموعة هاي الشبكة

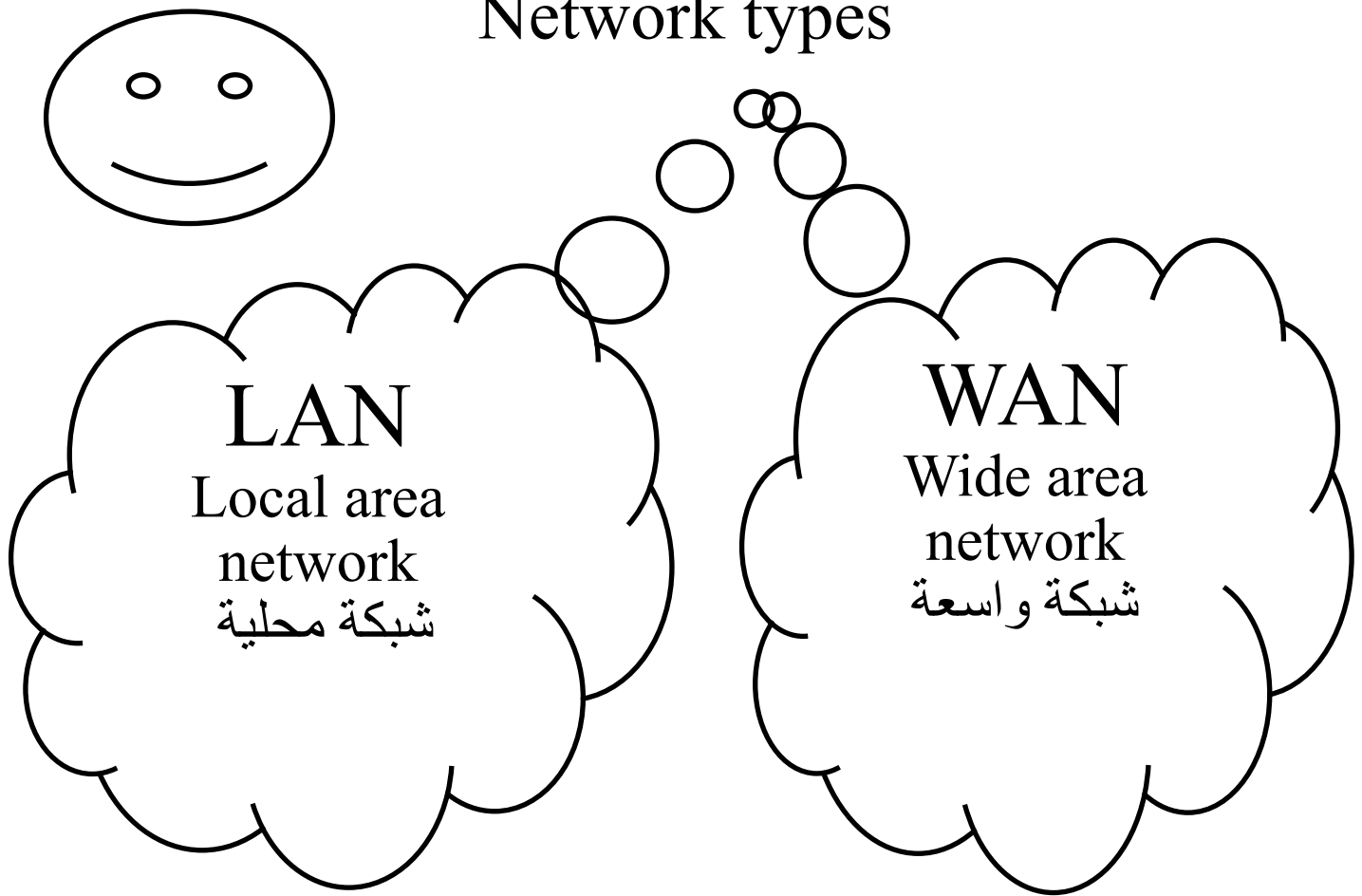
- Why do we need network?

حتى اعمل تواصل بين الاجهزة و منها بسهل علي تناقل البيانات و ارسل و استقبل بيانات و اعمل مشاركة بينهم لانه كلهم على شبكة وحدة متصلين

- **Devices** in the **same network** can easily **communicate** and **share resources** with each other.

- For example : you can control your TV with your mobile, or you can stream videos from your mobile to your TV.

Network types



LAN (local Area network)

- Is a **type of network**
- That has a **group of connected devices**
- Made up of —> **nearby computers**

من اهم خصائصها :

يتكون **الاجهزة قريبة** من بعض و عدد الاجهزة يكون كبير او صغير

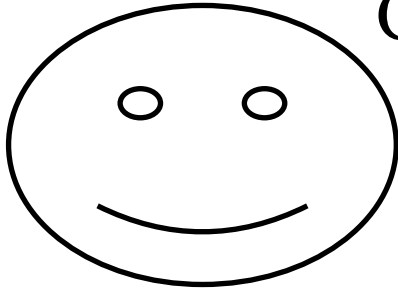
Ex —>

- **A LAN can have anywhere from a few devices** (like a home network) **to hundreds of devices** (like a school or hospital)
- **Large organizations** like colleges can **have multiple LANs** - maybe one per building.

كيف (computers) بتتفاهم مع بعض بعالم الشبكات؟؟

بناء على connection type

Communication on LAN —>
Connection type



Wired
اسلاك

- using wires or 'cables with

Ethernet

Ethernet

اسم الشبكة و معناها انه
(computers) كلها مشبوكة
مع بعض عن طريق اسلاك
بكونوا مشبوكين على جهاز
اسمه **switch**

Wireless

Without
wires
using
wi-fi

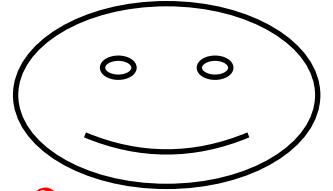
بتواصلوا مع
بعض عن
طريق

Router

Switch :-

هو جهاز بطلع منه سلك لكل (computers) مشبوك فيه
(computers يلي مشبوكين مع بعض)

Ethernet —>



Rules that computers follow for
communicating on a wired network

مجموعة من القواعد والقوانين يلي (computers) بتتبعها
حتى تتواصل مع بعض

Rules called protocol

Wireless (WIFI —> Wireless fidelity) :-

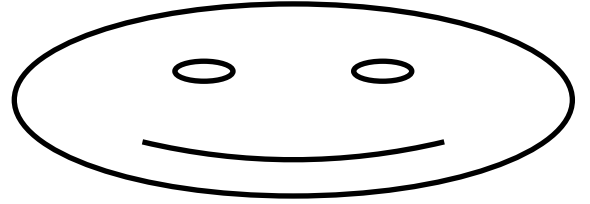
Rules that computers follow for
communicating through a radio signals

WI-FI (Wireless fidelity) —> is a popular way
to connect wirelessly to a network

. Wireless (WIFI > Wireless fidelity):-

هي عبارة عن مجموعة من القوانين يلي بتتبعها (computers) حتى تقدر تتواصل مع بعض من غير اسلاك و هي بتتواصل مع بعض عن طريق (radio signals) موجات الراديو

. protocol —>



Is a standard set of rules for how devices communicate through wires or wireless

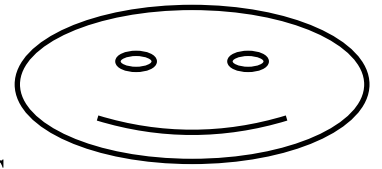
WAN (Wide Area Network)

. Is made of **connected LANs** that are **far from each other**

. **WANs connect thousand or even millions of devices** (like in different parts of the country or world)

. بتتكون من عدة (LANs) (شبكات local) عادة بستخدموها للبلاد و المسافات البعيدة

Homework



- 1 - what is the network ?

Is a group of connected computer and devices

- 2- Why do we need network?

Because Devices in the same network can easily communicate and share resources with each other

- 3- What is LAN vs WAN?

LAN —> (local area network) is **group of connected devices**, Made up of **nearby computers**

WAN —> (wide area network) Is made of **connected LANs** that are **far from each other**

- 4- How computers communicate together ?

Based on connection types (**wireless** or **Wired**)

- 5- What is Ethernet?

Rules that computers follow **for communicating on a wired network**

- 6- What is wireless?

Ans—> Rules that computers follow **for communicating** through a **radio signals**

- 7- What is protocol?

Ans—> Is a standard set of rules for how devices communicate through wires or wireless

- 8- What is WI-FI?

Ans—> Rules that computers follow **for communicating** through a **radio signals**

- 9- WI-FI stands for what ?

Ans—> Wireless fidelity